

End-behavior of a Kaplan-Meier Estimator for Repeated Events

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Abstract

Assuming a mixture cure model of customer return, we show that the Kaplan-Meier estimator applied to inter-visit gap times converges to the proportion of censored visits, regardless of the underlying rates of attrition.

1 Introduction

We want to model the following scenario. A restaurant tracks its customers' patterns of repeated visits. Data collection begins on day $d = 1$. On each day between day 1 and day x , the final day of data collection, some constant number, k , of new customers discover the restaurant and make their first visit. Thus, on day x , kx customers in total have visited the restaurant. To determine time to return, we will make independent and identically distributed draws from some probability distribution on the **positive** integers with finite moments (although we will only need finite fourth moment). It will be convenient for the number of these draws to be deterministic, depending only on x and not on the results of these draws. For each customer i , we make one draw T_{ij} for each positive integer j , representing the time to return *if the customer visits on day j* .

Note that if customer 1 has $T_{1,1} = 3$, then their next visit will be on day 4, and so $T_{1,2}$ and $T_{1,3}$ have no influence on their behavior at all.

Customer i visits the restaurant for the first time on day d_i . For each customer, we randomly select a day of attrition D_i , on the interval $[d_i, \infty)$ such that the CDF f_D of the distribution from which the D_i are drawn satisfies: $y_1 > y_2 \implies f_D(y_1|T_{ij}) \leq f_D(y_2|T_{ij})$. We do not assume that the D_i are independent of the T_{ij} .

The above model approximates a *mixture cure* model, since the “cured” population (who will never return to the restaurant) are modeled separately from the *at-risk* population who may still return.

2 The KM Estimator

The Kaplan-Meier estimator is the standard approach in oncology (and other medical disciplines) to estimating the time to some event (often called *survival time*) in the presence of censored data. The inputs to the estimator are times T_i and event/censoring indicators δ_i . The standard approach assumes one row per subject, with the δ_i indicating whether the event has occurred ($\delta_i = 1$) or not ($\delta_i = 0$) for each subject. Typically the estimator is written as a product of the proportion of subjects *at risk* at a given time who *survive*, or do not have the event. However, we can factor these terms as ratios $\frac{m-1}{m}$ by treating each event time as infinitesimally distinct. Suppose we have n records. Ordering the T_i in increasing order,

$$S(t) = \prod_{\{i: T_i \leq t\}} \left(\frac{n - (i + 1)}{n - i} \right)^{\delta_i}.$$

Suppose we take a snapshot of our data on day d . Keep in mind, the restaurant wouldn't be able to observe each of the T_{ij} - they would only observe the visits that actually occurred.

For each customer, let c_i be the time from their last visit to day d - the censoring time. Note that c_i is a function of x , of the T_{ij} , and of the D_i . If the customer discontinued at time D_i , their c_i would be the time from their last visit before time D_i .

On day x , we wish to estimate time-to-return. One plausible approach is to use the Kaplan-Meier estimator and to treat each visit as a separate observation. The final visits for each customer are the censored visits, and we have no deterministic way to distinguish among these censored visits which are due to true attrition and which correspond to a customer who will return but hasn't yet.

For each visit, all we know is the time to the next visit (if it occurred), and the censoring time if it did not. We also know whether each visit is censored. Each non-censored visit time is one of the T_{ij} , and the c_i are drawn from some distribution on $[d_i, x)$ which depends on the T_{ij} .

To simplify the analysis, we assume that we have chosen the distribution of the D_i so that the c_i are asymptotically uniform on $[0, x]$.

We re-index the non-censored visits in order of non-decreasing length as $\widehat{(T)}_\ell$ and we re-index the censoring time $\widehat{c}_i$ similarly.

We interleave the $\widehat{(T)}_\ell$ and $\widehat{c}_i$ with the standard convention that, in the case of ties, $\widehat{c}$ (the censored time) occurs after $\widehat{(T)}$ (the event time).

Call these times τ_i and let $\delta_i = 1$ if τ_i corresponds to an event $\widehat{T}_\ell$ or a censoring time $\widehat{c}$. We are finally ready to write down the formula for the KM estimator. Recall that kx is the total number of customers. Let V_x be the total number of visits which have occurred by time x . Note that V_x is bounded above by kx^2 . Then

$$S_x(t) = \prod_{\{i: \tau_i = t\}} \left(\frac{V_x - (i + 1)}{V_x - i} \right)^{\delta_i}.$$

Really, we are only interested in the final value $S_x(x)$, which we abbreviate as S_x :

$$S_x = \prod_{i=1}^{V_x} \left(\frac{V_x - (i + 1)}{V_x - i} \right)^{\delta_i}.$$

Our main result is:

Theorem 1. *With $\widehat{c}_i$ uniformly distributed in $[0, x]$, $\widehat{T}_\ell$ i.i.d. as above, and τ_i defined as above, let $V_x = V_e + V_c$, where V_e is the number of $\widehat{T}_\ell$ and V_c the number of $\widehat{c}_i$. Then, with probability 1,*

$$\lim_{x \rightarrow \infty} S_x \cdot \frac{V_x}{V_c} = 1.$$

3 Bounds for the asymptote of the KM estimator

Note that $S_x \leq \frac{V_c}{V_x}$, because if all events occur before all censored records, the estimator simply becomes:

$$\frac{V_x - 1}{V_x} \cdot \frac{V_x - 2}{V_x - 1} \cdots \frac{V_x - V_e}{V_x - V_e + 1} = \frac{V_c}{V_x}.$$

and that any other value of S_x would replace some of the terms of this product with smaller values.

Thus, we wish to find a lower bound for S_x with probability 1 and which converges to $\frac{V_c}{V_x}$ in the limit (i.e., the quotient of the two converges to 1 in the limit). Note that by our assumptions of how the customers originate (k per day), we have $V_c = kx$ and we have $V_x < kx^2$.

We will argue that relatively few of the censored times occur before the final uncensored time. Let b_x be the number of such censored times.

We will construct a function $\varphi(x)$ such that $b_x < \varphi(x)$ with probability 1 and such that $\varphi(x)/x$ approaches 0 in the limit. Assuming that φ exists, we can prove the theorem:

Proof. As above,

$$S_x = \prod_{i=1}^{V_x} \left(\frac{V_x - (i+1)}{V_x - i} \right)^{\delta_i}.$$

we imagine rearranging the δ_i so that the first b_x terms are 1. This replaces some terms of the product with smaller values. Therefore,

$$S_x \geq \frac{V_x - b_x - 1}{V_x - b_x} \cdots \frac{V_x - V_c - b_x - 1}{V_x - V_c - b_x} = \frac{V_c - b_x - 1}{V_x - b_x}.$$

Therefore:

$$\lim_{x \rightarrow \infty} S_x \frac{V_c}{V_x} \leq \lim_{x \rightarrow \infty} \frac{V_c - b_x - 1}{V_x - b_x} \cdot \frac{V_c}{V_x} \quad (1)$$

$$\leq \lim_{x \rightarrow \infty} \frac{V_c - b_x - 1}{V_c} \quad (2)$$

$$\leq \lim_{x \rightarrow \infty} \frac{V_c - \varphi(x) - 1}{V_c} \quad (3)$$

$$= \lim_{x \rightarrow \infty} \frac{kx - \varphi(x) - 1}{kx} \quad (4)$$

$$= \lim_{x \rightarrow \infty} 1 - \frac{-\varphi(x)}{kx} = 1 \quad (5)$$

as desired. $\square$ $\square$

Thus, all that remains is to choose an appropriate $\varphi(x)$. For the above proof to hold, we need $\varphi(x) < o(x)$. To get $b_x < \varphi(x)$ with probability 1, we will need $\varphi(x) > o(\sqrt{x})$. Any such function will do. It remains to show that this lower bound on $\varphi(x)$ is sufficient to get the bound on b_x as x approaches infinity.

Recall that b_x is the number of $\hat{c}_i$ which occur before the maximum of the $\hat{T}_\ell$. However the number of $\hat{T}_\ell$ depends on the $\hat{T}_\ell$ and $\hat{c}_i$ in a complicated manner. To simplify the analysis, we instead consider $\star b_x$, the number of $\hat{c}_i$ less than or equal to the maximum of the precisely kx^2 i.i.d T_{ij} . We drop the double index on the T as we no longer need to keep track of the customer to which each interval belongs.

We use the following fact (verifiable with straightforward calculus) without proof.

Fact 2. *If $k > 1$ and $x \in [1, \infty)$, then*

$$\left(1 - \frac{1}{kx^4} \right)^{kx^2} > 1 - \frac{1}{x}.$$

We'll need two lemmas:

Lemma 3.

$$\lim_{x \rightarrow \infty} \sqrt{x} P(\max(T_j) > \sqrt[4]{x}) = 0.$$

Proof. Note that the maximum is taken over kx^2 i.i.d. T_j . For any T with this distribution, recall that T has finite m th moment for all m , so we have:

$$\lim_{t \rightarrow \infty} t^m P(T > t) = 0.$$

We need the particular case where $t = \sqrt[4]{x}$ and $m = 16$. Multiplying by the constant k , we have:

$$\lim_{t \rightarrow \infty} kx^4 P(T > \sqrt[4]{x}) = 0.$$

In particular, for x sufficiently large,

$$kx^4 P(T > \sqrt[4]{x}) < 1.$$

or, suggestively, by Fact 2 above:

$$P(T > \sqrt[4]{x}) < \frac{1}{kx^4}.$$

Applying this to the fact, we shrink $\frac{1}{kx^4}$ to $P(T > \sqrt[4]{x})$, increasing the LHS of the inequality, so

$$(1 - P(T > \sqrt[4]{x}))^{kx^2} > 1 - \frac{1}{x}.$$

Rearranging, for x sufficiently large, we have what we'll eventually need:

$$1 - (1 - P(T > \sqrt[4]{x}))^{kx^2} < \frac{1}{x}. \quad (6)$$

Because the T_j are independent, we have

$$P(\max(T_j) > \sqrt[4]{x}) = 1 - (P(T_j \leq \sqrt[4]{x}))^{kx^2}.$$

For simplicity, we drop the subscript and refer to a random variable T with the same distribution as the T_j . Then,

$$\lim_{x \rightarrow \infty} \sqrt{x} P(\max(T_j) > \sqrt[4]{x}) = \lim_{x \rightarrow \infty} \sqrt{x} \left(1 - (1 - P(T > \sqrt[4]{x}))^{kx^2} \right).$$

But by equation 6, we now have (for x sufficiently large):

$$\lim_{x \rightarrow \infty} \sqrt{x} P(\max(T_j) > \sqrt[4]{x}) < \lim_{x \rightarrow \infty} \sqrt{x} \cdot \frac{1}{x} = 0.$$

as desired. □

With this result, we no longer need to distinguish the T_j , we only care about their maximum, which we denote with the random variable $T_{\max}$.

Lemma 4. $\lim_{x \rightarrow \infty} \sqrt{x} \cdot P(c_i < T_{\max}) = 0.$

Proof. Note that by our assumption that the c_i are uniformly distributed (this is the only place we use this assumption, and it is easy to see how we might weaken it), for any N ,

$$P(c_i < N) = \frac{N}{X}.$$

We will bound $\sqrt{x} \cdot P(c_i < T_{\max})$ above. Decompose $P(c_i < T_{\max})$ conditionally on $T_{\max}$ at the boundary $\sqrt[4]{x}$.

$$P(c_i < T_{\max}) = P(T_{\max} \leq \sqrt[4]{x}) \cdot P(c_i < T_{\max} \mid T_{\max} \leq \sqrt[4]{x}) \quad (7)$$

$$+ P(T_{\max} > \sqrt[4]{x}) \cdot P(c_i < T_{\max} \mid T_{\max} > \sqrt[4]{x}) \quad (8)$$

$$\leq P(T_{\max} \leq \sqrt[4]{x}) \cdot P(c_i < \sqrt[4]{x}) + P(T_{\max} > \sqrt[4]{x}) \quad (9)$$

$$= P(T_{\max} \leq \sqrt[4]{x}) \cdot \frac{\sqrt[4]{x}}{x} + P(T_{\max} > \sqrt[4]{x}) \quad (10)$$

$$\leq \frac{\sqrt[4]{x}}{x} + P(T_{\max} > \sqrt[4]{x}) \quad (11)$$

. Now, we take the limit as x approaches infinity and apply Lemma 3.

$$\lim_{x \rightarrow \infty} \sqrt{x} \cdot P(c_i < T_{\max}) \leq \lim_{x \rightarrow \infty} \sqrt{x} \cdot \left(\frac{\sqrt[4]{x}}{x} + P(T_{\max} > \sqrt[4]{x}) \right) \quad (12)$$

$$= \lim_{x \rightarrow \infty} \frac{1}{\sqrt[4]{x}} + \sqrt{x} P(T_{\max} > \sqrt[4]{x}) \quad (13)$$

$$= 0 \quad (14)$$

. As desired, $\lim_{x \rightarrow \infty} \sqrt{x} \cdot P(c_i < T_{\max}) = 0$. □

Finally, we're ready to prove our result.

Proposition 5. *Suppose $\varphi(x) > o(\sqrt{x})$. Then*

$$\lim_{x \rightarrow \infty} P(b_x < \varphi(x)) = 1.$$

Proof. Let $\pi(x) = P(c_i > T_{\max})$. Then by Lemma 4, $\lim_{x \rightarrow \infty} \sqrt{x}\pi(x) = 0$. Moreover, note that b_x is the sum of kx independent Bernoulli-distributed variables with $p = \pi(x)$, and so it has distribution $\text{Binom}(kx, \pi(x))$. By Markov's inequality:

$$P(b_x \geq \varphi(x)) \leq \frac{kx\pi(x)}{\varphi(x)}.$$

Hence,

$$P(b_x < \varphi(x)) \geq 1 - \frac{kx\pi(x)}{\varphi(x)}.$$

Finally, taking the limit:

$$\lim_{x \rightarrow \infty} P(b_x < \varphi(x)) \geq 1 - \lim_{x \rightarrow \infty} \frac{kx\pi(x)}{\varphi(x)} \quad (15)$$

$$\geq 1 - \lim_{x \rightarrow \infty} k\sqrt{x}\pi(x) \frac{\sqrt{x}}{\varphi(x)} \quad (16)$$

. By Lemma 4 and because $\varphi(x) > o(\sqrt{x})$, this is equal to 1. □